Is Equity Return Affected by Country or by Industry?

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Abstract

Country and industry affiliation are commonly used to account for the relative performance of stocks. These two factors usually co-moves and mutually contributes to the equity returns. This paper extracts the effect of 15 country and 11 industry dynamics and calculates the relative importance of them on the stock return volatility. Results are presented as ratios, which is country or industry related volatility to the total volatility. In the latest data, country and industry dynamics both account for half of the volatility. The narrowing gap in the past twenty years between these two factors implies the achievement of globalization.

Keywords: Volatility Decomposition, Exchange Rate, Emerging Market, Diversification

Contents

1. Introduction

In the development of financial market, there have occurred various forces driving the movement of stock price. During the 19th century, the United Kingdom led the Industrial Revolution, which brought most people from agricultural activities to industrial activities. Textile manufacturers, Iron and railroad companies emerged since then. After entering into the 20th century, new technologies are invented more and more quickly. As a comparison, only about 100,000 patents were issued in the 19th century, but over 5 million patents were issued in the 20th century. Different eras have different leading industries, and these industries may also get support from the central governments. Country and industry evolve to be two big hands on the stock market. Understanding the co-movement helps us to manage risk and construct a better global stock portfolio.

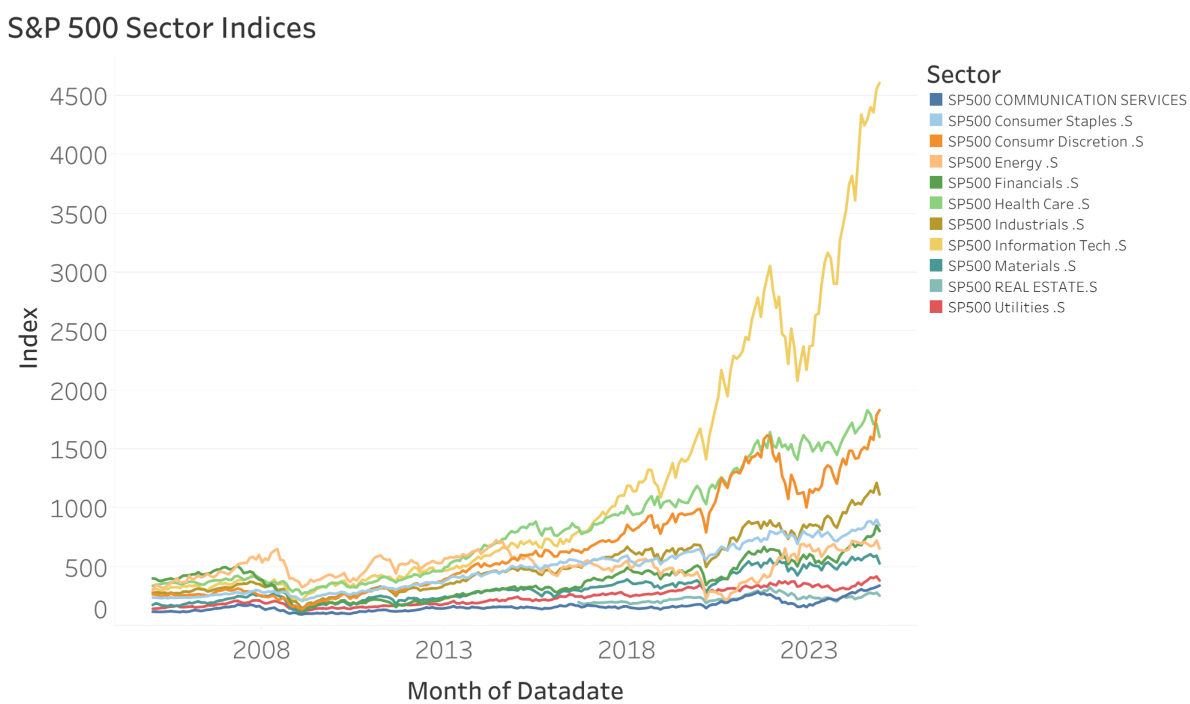

Figure 1: S&P 500 sectors indice

Figure 1 plots the time series trend of different S&P 500 sectors. The Information technology sector has shown the most dramatic growth, especially since around 2015, surging steeply and far outpacing all other sectors by 2024. Consumer discretionary and health care also experienced strong growth post-2010, but with more volatility. This represents the rise of information technology.

Beckers et al. (1996) found that in EU, country factors outperformed industry factors. The worldwide model has stronger country factors and smaller industry influences. Table 1 lists the results from the Beckers' paper. In EU, the average global factor variance is 0.0022, which explains the largest percentage of volatility. The second largest is country factors, while industry factors only explain 0.0006. By dividing the explanatory power into three parts, the global market factor alone explains 21.07 percent of volatility, industry adds 4.30 percent explanatory power, whereas the countries add 14.33 percent to the global industries-only model.

Sample	Variance of Global Market Factor	Average Variance of Country Factors	Average Variance of Industry Factors
Worldwide	0.001764	0.0015306	0.0002769
EU-only	0.002209	0.0012089	0.0006074

Table 1: Average Variance without October 1987

However, Baca et al. (2000) concluded that the spread between country-component and sector-component variation has narrowed so sharply in recent years that it is now statistically insignificant. Figure 2 shows the gap between the average variance of the pure country return components and pure sector return components over 20 years, from 48-month time period March 31,1979 to 1999. At the beginning of 1990s, the ratio of average country coefficient variance to average sector coefficient variance was 3.10. While in the period of 1995-1999, this ratio has dropped to 1.23.

In addition, Gerard et al. (2005) thought that country-based strategies perform at least as well as industry-based strategies. Serra (2000) included emerging market data and found that emerging markets' indices are driven by country factors, and almost not by the industrial composition of the indices. Then it comes to the motivation of this research: is equity return affected more by country or industry, and to what extent?

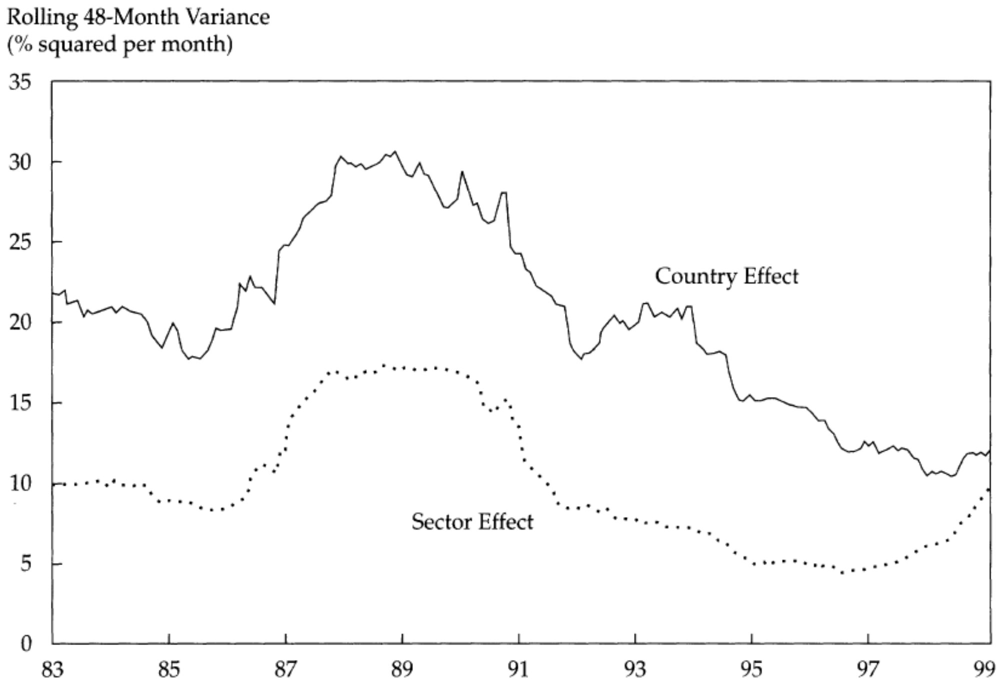

Figure 2: Relative Importance of Countries and Sectors over Time: Rolling 48-Month Average Effect Variances (source: Baca et al. (2000))

This paper is composed of five parts. Section 2 describes the data being used in the following research. Section 3 introduces the models. The empirical part of this paper has two steps. In the first step, a weighted least square regression will be applied to get the influence of country and industry effects on firm stock return. The next step is decomposition of firm stock return volatility. Aim of this step is to know how much volatility can be explained by country and industry effects. Section 4 analyzes the results got from the models in section 3. Then a robustness check is launched to test the validity of the regression. Section 5 concludes all findings in this paper and proposes some plausible causes of the phenomenon.

2. Data

The data used in this paper is obtained from Compustat. It covers monthly financial information of over 30,000 firms, from 15 countries and 11 industries. The time period is 20 years from January 2005 to December 2024. The 15 countries are Australia, Belgium, Canada, China, Hong Kong, Denmark, France, Germany, Ireland, Italy, Japan, Netherlands, Switzerland, the UK and the USA. Global Industry Classification Standard (GICS) is applied in this paper, so the 11 industries are energy, materials, industrials, consumer discretionary, consumer staples, health care, financials, information technology, communication services, utilities, and real estate. This classification consists of not only traditional industries, but also new ones like informational technology and Equity Real Estate Investment Trusts (REITs).

Despite that only US and Canada have data in 2005, all countries have full firm information starting from January 2006. Since then, no country is represented by fewer than 32 firms (Ireland in 2006). Over half of all countries have more than 500 firms in every year. Large economies such as Japan and the US, have about 3000 and 6000 firms in 2006, but this number declines over the time period till 2023. In 2023, there are only 3103 and 4659 firms. One of the possible reasons is that small firms are acquired by large firms.

Compared with previous studies, this paper includes the Chinese stock market in the research for the first time. Since this is an emerging market in this century, some of the company performances may be distinct, such as the equity short-term reversal anomaly. This is the most significant anomaly in the Chinese stock market. Stocks which previously had a higher return have a lower expected return in the future.

There is no market capitalization data directly from the Compustat database. It is calculated as shares outstanding multiplied by the stock price at the end of each month. Exchange rate is used to convert all foreign currencies to US dollars, in order to further get the weight of every country and industry.

Table 1 is a summary of all country returns. Among all countries, the US has the highest yearly return of 1.2%, followed by Australia and China of 1.1%. The average yearly return of all countries is 0.8%. Some of them have a relatively low return. For example, Italy, Ireland and Denmark have 0.3%, 0.4% and 0.4%. One common thing among them is that they are all well-developed countries. The stock market is mature and does not generate too much returns.

Country portfolios	**Mean**	**S.D.**	**Skewness**	**Kurtosis**
Australia	0.011	0.269	17.361	678.914
Belgium	0.006	0.112	3.824	81.147
Canada	0.016	0.310	18.162	734.865
Switzerland	0.005	0.178	74.852	7952.074
China	0.011	0.135	2.027	25.249
Germany	0.006	0.107	2.289	46.945
Denmark	0.004	0.099	-0.010	12.881
France	0.007	0.167	45.797	3791.695
UK	0.006	0.167	38.669	4121.134
Hong Kong	0.010	0.250	35.769	2033.716
Ireland	0.004	0.153	2.574	30.207
Italy	0.003	0.194	47.820	3621.608
Japan	0.006	0.162	28.334	1423.308
Netherlands	0.006	0.149	26.328	1697.958
US	0.012	0.264	22.556	1080.126
Average	0.008	0.181	24.423	1822.122

Table 2: Mean Return by Country

Industry portfolios	**Mean**	**S.D.**	**Skewness**	**Kurtosis**
Energy	0.011	0.266	19.553	910.121
Materials	0.012	0.242	19.606	913.637
Industrials	0.009	0.179	23.257	1320.336
Consumer Discretionary	0.007	0.197	25.306	1590.124
Consumer Staples	0.008	0.192	34.023	2360.365
Health Care	0.012	0.303	20.452	868.173
Financials	0.006	0.198	43.806	3123.979
Information Technology	0.013	0.258	21.090	898.911
Communication Services	0.008	0.279	30.362	1635.742
Utilities	0.005	0.152	45.698	3973.334
Real Estate	0.011	0.272	31.703	1618.623
Average	0.009	0.231	28.623	1746.668

Table 3: Mean Return by Industry

The industrial level summary statistics is listed in table 2. The difference among various industries is less than that among all countries. The highest return industry is information technology. This is in line with the trend in the past twenty years, with the surge of many IT companies like Apple, Microsoft and Nvidia. The average return of all industries is 0.9%. One thing to note is that, compared with Catao and Timmerman's paper in 2003, their standard deviation ranged from 2.77 to 7.28 for country portfolios, and from 1.61 to 4.34 for industry portfolios. Their data time period is from February 1973 to February 2002, right before my time period. This implies a sort of lower volatility of stock returns in recent years.

3. Methodology

3.1 Country and Industry effects on Return

This paper will follow Heston and Rouwenhorst (1994, 1995) to extract the return attributed to country and industry dynamics separately:

$$R_{ijkt} = \alpha_t + \beta_{jt} + \gamma_{kt} + \varepsilon_{it} \tag{1}$$

where R_{ijkt} is return at time t of the i^{th} firm which belongs to the j^{th} industry and the k^{th} country. α_t is the global factor, which is a common factor for all firms. This captures the common effect influencing every firm in the world in a certain time period. β_{jt} is the excess return owing to industry j. γ_{kt} is the excess return owing to country k. ε_{it} is the firm-specific error term.

$$R_{ijkt} = \alpha_t + \sum_{j=1}^{J} e_{ij\beta}\beta_{jt} + \sum_{k=1}^{K} e_{ij\gamma}\gamma_{kt} + \varepsilon_{it} \tag{2}$$

Now assume that there are J industries and K countries. Here in the data sample, $J = 11$ and $K = 15$. Equation (1) can be rewritten as (2). $e_{ij\beta}$ and $e_{ij\gamma}$ are two dummy variables.

$e_{ij\beta} = 1$ if firm i is in industry j and 0 otherwise. $e_{ij\gamma} = 1$ if firm i is in country k and 0 otherwise. In this way, it is easier to put β_{jt} and γ_{kt} in the regression model. The two dummy variables are orthogonal and therefore they build a perfect factor model. $e_{ij\beta}$ and $e_{ij\gamma}$ can now represent two completely different driving force on the movement of firm equity return. For every firm or every R_{ijkt}, their $e_{ij\beta}$ and $e_{ij\gamma}$ can be 1 only once because each firm only belongs to one country and one industry. However, equation (2) has the problem of multicollinearity. The reason is that, the sum of all $e'_{ij\beta}s$ or all $e'_{ij\gamma}s$ is always 1. To solve this problem, some restrictions must be applied to the regression:

$$\begin{aligned} \sum_{j=1}^{J} \beta_j \sum_{i=1}^{N} e_{ij\beta} x_i = \sum_{j=1}^{J} \beta_j w_j = 0 \\ \sum_{k=1}^{K} \gamma_k \sum_{i=1}^{N} e_{ij\gamma} x_i = \sum_{k=1}^{K} \gamma_k v_k = 0 \end{aligned} \tag{3}$$

where N is the total number of firms, x_i is the beginning share of the stock in the global market capitalization. w_j is the share of market capitalization of industry j in the global market, while v_k is the share of market capitalization of country k in the global market. The sum of all country and industry effects times their weight is zero. With the restrictions, the regression has been turned into a weighted least square regression. One of the industry and country needs to be omitted to avoid collinearity. In the regression, country of Australia and industry of energy are dropped in the weighted least regression. Their effect on the equity return can be got as the additive inverse of the sum of other effects, according to the restrictions (3).

Therefore, the equation can be written in the matrix form:

$$R_{ijkt} = \alpha_t + e'_{i\beta}\beta_{jt} + e'_{i\gamma}\gamma_{kt} + \varepsilon_{it} \tag{4}$$

where $e'_{i\beta}$ is a $J \times 1$ dummy vector indicating whether a firm is in a specific industry and $e'_{i\gamma}$ is a $K \times 1$ dummy vector indicating whether a firm is in a specific country.

3.2 Volatility Analysis

In the next step, this paper is going to figure out to what extent the industry and country dynamics influence the equity return volatility. In order to do this, the first step is to decompose return into two parts, using the model developed by Griffin and Karolyi (1998):

$$R_{kt} - \hat{\alpha}_t = \sum_{j=1}^{J} \omega_{jkt}^{\beta} \hat{\beta}_{jt} + \hat{\gamma}_{kt} \tag{5}$$

$$R_{jt} - \hat{\alpha}_t = \sum_{k=1}^{K} \omega_{jkt}^{\gamma} \hat{\gamma}_{kt} + \hat{\beta}_{jt} \tag{6}$$

Here in equation (5) and (6), the left hand side eliminates the effect which is common to all firms. The right hand side of equation (5) decomposes the return into two parts: weighted industry effect plus an average country effect. Similarly, equation (6) decomposes the return into weighted country effect plus an average industry effect. Additionally, we can get the variance of the excess returns:

$$\begin{aligned} Var(R_{kt} - \hat{\alpha}_t) &= (\omega_{kt}^{\beta})' \Omega_{\beta} \omega_{kt}^{\beta} + e_k' \Omega_{\gamma} e_k + 2(\omega_{kt}^{\beta})' Cov(\beta_{jt}, \gamma_{kt}) \\ Var(R_{jt} - \hat{\alpha}_t) &= (\omega_{jt}^{\gamma})' \Omega_{\gamma} \omega_{jt}^{\gamma} + e_j' \Omega_{\beta} e_k + 2(\omega_{jt}^{\gamma})' Cov(\beta_{jt}, \gamma_{kt}) \end{aligned} \tag{7}$$

where ω_{kt}^{β} is the market capitalization weights of country k, and ω_{jt}^{γ} is the market capitalization weights of industry j. Ω_{β} is the variance of the parameter estimate $\hat{\beta}_{jt}$ and Ω_{γ} is the variance of the parameter estimate $\hat{\gamma}_{kt}$. Then the percentages of volatility caused by industry or country effects will be calculated.

4. Results

4.1 Equity Return Influencing Factors

This part reports some empirical results of the regressions. Table 4 shows the result of panel weighted least square regression. Most coefficients are significant. From the table, the global factor on the equity return is 2.3%. Among the country effects, Belgium, Canada, Switzerland, China, Germany, France, the UK, Ireland, Italy, Japan, Netherlands, and the US are significant. Only Denmark and Hong Kong are insignificant. About half of the countries have a negative effect on the stock returns. Belgium has the strongest negative effect of -6.3%, while Switzerland has the strongest positive effect of 8.1%.

A possible reason is that Belgium's stock market is heavily weighted toward legacy industries like chemicals, industrials, and financials. In the past score, the industry with greatest growth was information technology. These sectors tend to underperform in periods dominated by tech-led global growth. Besides, Belgium is short of globally dominant technology or innovation-driven companies. Its listed firms are typically mid-sized, low-volatility exporters rather than aggressive growth stocks, which underperform in bull markets. On contrary, Switzerland is home to world-leading multinational companies like Nestlé, Roche, and Novartis. These firms operate in defensive sectors such as consumer staples and healthcare, which tend to be more resilient during global downturns. Their stable earnings and global revenue bases helped Swiss stocks remain attractive throughout market cycles. By the way, as a special currency in the European area, Swiss franc is widely considered a safe-haven currency, especially in times of economic or geopolitical stress. Having an independent currency enables Switzerland to publish better fiscal and monetary policies which serves its country better. As a result, Swiss assets—including equities—often benefit from capital inflows during periods of global uncertainty. In addition, Switzerland's stable political system and low public debt bolster investor confidence.

Variable	Coefficient	Std. Error	t-stat	p-value	95% Confidence Interval
α	0.025***	0.001	19.380	0.000	[0.022, 0.027]
Belgium	-0.063***	0.003	-19.107	0.000	[-0.069, -0.056]
Canada	-0.005***	0.002	-2.865	0.004	[-0.008, -0.002]
Switzerland	0.081***	0.003	32.122	0.000	[0.076, 0.086]
China	-0.011***	0.001	-11.682	0.000	[-0.013, -0.009]
Germany	-0.012***	0.002	-7.681	0.000	[-0.015, -0.009]
Denmark	-0.002	0.003	-0.617	0.537	[-0.008, 0.004]
France	-0.012***	0.001	-7.893	0.000	[-0.015, -0.009]
UK	0.022***	0.001	17.831	0.000	[0.020, 0.024]
Hong Kong	0.002	0.003	0.519	0.604	[-0.004, 0.007]
Ireland	0.016***	0.006	2.657	0.008	[0.004, 0.027]
Italy	0.045***	0.002	21.984	0.000	[0.041, 0.049]
Japan	-0.019***	0.001	-19.444	0.000	[-0.021, -0.017]
Netherlands	-0.019***	0.003	-6.050	0.000	[-0.025, -0.013]
US	0.003***	0.001	3.160	0.002	[0.001, 0.005]
Materials	-0.003**	0.001	-2.532	0.011	[-0.006, -0.001]
Industrials	0.003**	0.001	2.196	0.028	[0.000, 0.005]
Consumer Discretionary	0.003***	0.001	2.647	0.008	[0.001, 0.006]
Consumer Staples	0.008***	0.001	5.305	0.000	[0.005, 0.011]
Health Care	-0.011***	0.001	-8.776	0.000	[-0.014, -0.009]
Financials	0.002*	0.001	1.646	0.100	[-0.000, 0.005]
Information Technology	-0.002*	0.001	-1.696	0.090	[-0.005, 0.000]
Communication Services	0.004***	0.002	2.593	0.010	[0.001, 0.007]
Utilities	-0.015***	0.002	-7.708	0.000	[-0.018, -0.011]
Real Estate	-0.015***	0.002	-7.776	0.000	[-0.018, -0.011]

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table 4: Weighted Least Square Regression Results

As comparison, more than half of the industry effects are significant. Consumer staples has the strongest effect of 1.0%. One of the main focus, information technology, has a negative effect here. A plausible explanation is the substitution effect. Some other variables may have taken part of the effect of the industries.

4.2 Variation of Volatility

This part reports the volatility decomposition. From the equity return data obtained in the parts above, we get a summary table of the volatility of different countries and industries:

Country	**Volatility**
Australia	0.245
Belgium	0.112
Canada	0.219
Switzerland	0.245
China	0.133
Germany	0.124
Denmark	0.143
France	0.137
UK	0.217
Hong Kong	0.188
Ireland	0.248
Italy	0.225
Japan	0.107
Netherlands	0.153
US	0.196
Mean	**0.186**

Table 5: Firm volatility of different countries

Table 5 and Table 6 list the firm volatility of different countries and industries. Over the past twenty years, Ireland stocks had the highest volatility of 0.248, while Japan had the lowest volatility of 0.107. During mid-2000s, Ireland economy experienced a rapid growth. The average growth rate was 9.4% between 1995 and 2000, and 5.9% during the following decade until 2008. The fluctuation of euro to pound sterling and US dollar greatly influenced Ireland's export of pharmaceutical products. During the 2008 recession, Irish property prices collapsed but just four years later, its economy recovered strongly. Due to the local corporate tax, many global firms have large Irish operations. The small market size also makes the economy of Ireland more volatile. Ó Gráda and O'Rourke (2022) think that one potential cause is generally pro-cyclical fiscal policy. The another is that complementary labor and capital flows into and out of the country, which occurred over the course of the cycle.

At the same time, Japanese firms tend to be conservatively managed, with strong balance sheets and large cash reserves which contribute to more stable stock performance. From the start of new century, after the appreciation of Japanese Yen, its long-standing low inflation and aging population have led to slower economic growth expectations and lower investor risk appetite—both factors suppress volatility.

Industry	**Volatility**
Energy	0.230
Materials	0.190
Industrials	0.147
Consumer Discretionary	0.160
Consumer Staples	0.157
Health Care	0.203
Financials	0.153
Information Technology	0.169
Communication Services	0.193
Utilities	0.128
Real Estate	0.157
Mean	**0.172**

Table 6: Firm volatility of different industries

Among all industries, energy volatiled mostly at 0.230. The energy sector is highly sensitive to geopolitical events, such as conflicts in oil-rich regions like Middle East. Wars, sanctions, and political instability can disrupt supply chains and lead to sudden price swings. Global demand for oil and energy products fluctuates based on economic growth and technological advancements. OPEC countries usually increase or reduce their productivity, which is a great shock to the energy price. Table 7 and Table 8 provide more details for the distribution of firm and industry level volatility.

Table 9 presents the decomposition of firm volatility into country effect and industry effect. The number shows the average proportion of firm volatility which can be explained by country volatility and industry volatility, from 2005 to 2024. Both columns are ratios of country-related volatility and industry-related volatility to the total volatility. Switzerland is the country with greatest country effect of 72.7%. China and Japan have the greatest industry effect of 70.8% and 71.1% respectively.

Table 10 shows the time series trend of the two parts of volatility. Before 2005, country factor accounted for 52% of total volatility. After that, this number first lowered and then increased. According to the latest data, country effect and industry effect almost have the

Year	Australia	Belgium	Canada	Switzerland	China	Germany	Denmark	France	UK	Hong Kong	Ireland	Italy	Japan	Netherlands	US
2005			0.192												0.220
2006	0.230	0.070	0.190	0.266	0.140	0.118	0.098	0.147	0.225	0.141	0.595	0.282	0.088	0.280	0.186
2007	0.214	0.079	0.182	0.310	0.216	0.121	0.096	0.143	0.219	0.217	0.540	0.258	0.093	0.279	0.174
2008	0.257	0.113	0.259	0.314	0.191	0.156	0.146	0.190	0.229	0.224	0.432	0.277	0.130	0.224	0.221
2009	0.315	0.144	0.307	0.261	0.144	0.170	0.178	0.179	0.261	0.226	0.269	0.295	0.128	0.178	0.274
2010	0.253	0.109	0.177	0.229	0.121	0.127	0.142	0.105	0.199	0.175	0.164	0.270	0.104	0.185	0.200
2011	0.231	0.195	0.217	0.207	0.107	0.113	0.126	0.129	0.177	0.201	0.270	0.295	0.094	0.150	0.179
2012	0.255	0.132	0.180	0.211	0.119	0.110	0.127	0.127	0.168	0.157	0.300	0.343	0.100	0.118	0.182
2013	0.262	0.090	0.210	0.205	0.127	0.108	0.138	0.108	0.207	0.159	0.201	0.291	0.120	0.117	0.183
2014	0.270	0.097	0.203	0.206	0.118	0.103	0.104	0.123	0.204	0.188	0.176	0.213	0.090	0.123	0.171
2015	0.280	0.101	0.247	0.219	0.233	0.123	0.116	0.119	0.212	0.213	0.173	0.181	0.094	0.104	0.177
2016	0.278	0.106	0.290	0.205	0.151	0.111	0.092	0.122	0.208	0.157	0.228	0.156	0.099	0.105	0.187
2017	0.228	0.093	0.224	0.199	0.100	0.112	0.080	0.109	0.199	0.170	0.077	0.236	0.086	0.095	0.167
2018	0.196	0.118	0.202	0.202	0.109	0.113	0.090	0.109	0.197	0.150	0.087	0.227	0.163	0.108	0.162
2019	0.219	0.111	0.220	0.209	0.122	0.110	0.152	0.131	0.223	0.165	0.112	0.231	0.104	0.097	0.177
2020	0.314	0.154	0.324	0.287	0.131	0.168	0.236	0.209	0.320	0.192	0.153	0.176	0.138	0.150	0.263
2021	0.224	0.094	0.198	0.207	0.140	0.119	0.230	0.125	0.204	0.252	0.110	0.130	0.102	0.111	0.205
2022	0.207	0.104	0.203	0.289	0.137	0.125	0.206	0.134	0.225	0.221	0.121	0.142	0.091	0.109	0.201
2023	0.201	0.099	0.211	0.279	0.095	0.127	0.171	0.154	0.237	0.196	0.118	0.121	0.098	0.125	0.209
2024	0.222	0.072	0.146	0.399	0.150	0.096	0.184	0.119	0.191	0.101	0.075	0.354	0.107	0.088	0.151
Mean	0.245	0.112	0.219	0.245	0.133	0.124	0.143	0.137	0.217	0.188	0.248	0.225	0.107	0.153	0.196

Table 7: Firm volatility by country and year

Year	Energy	Materials	Industrials	Consumer Discretionary	Consumer Staples	Health Care	Financials	Information Technology	Communication Services	Utilities	Real Estate
2005	0.240	0.260	0.200	0.204	0.181	0.211	0.182	0.211	0.333	0.146	0.432
2006	0.198	0.180	0.142	0.168	0.144	0.180	0.182	0.171	0.242	0.126	0.171
2007	0.201	0.185	0.143	0.165	0.168	0.179	0.178	0.164	0.215	0.155	0.154
2008	0.256	0.213	0.188	0.201	0.167	0.212	0.223	0.197	0.199	0.179	0.195
2009	0.287	0.246	0.192	0.224	0.204	0.257	0.234	0.225	0.241	0.138	0.241
2010	0.248	0.202	0.135	0.163	0.147	0.186	0.179	0.166	0.199	0.134	0.153
2011	0.231	0.160	0.131	0.150	0.137	0.168	0.168	0.153	0.192	0.113	0.136
2012	0.200	0.174	0.135	0.150	0.156	0.185	0.154	0.154	0.225	0.126	0.125
2013	0.204	0.188	0.141	0.154	0.152	0.202	0.141	0.173	0.221	0.122	0.129
2014	0.208	0.188	0.131	0.135	0.149	0.204	0.123	0.150	0.174	0.106	0.121
2015	0.274	0.223	0.162	0.149	0.168	0.212	0.123	0.183	0.182	0.126	0.153
2016	0.276	0.219	0.148	0.141	0.147	0.202	0.137	0.157	0.154	0.135	0.145
2017	0.230	0.164	0.123	0.122	0.125	0.196	0.109	0.147	0.132	0.084	0.113
2018	0.220	0.165	0.147	0.142	0.149	0.189	0.101	0.161	0.139	0.104	0.129
2019	0.234	0.171	0.137	0.136	0.146	0.203	0.096	0.158	0.157	0.097	0.130
2020	0.302	0.225	0.179	0.197	0.193	0.263	0.181	0.198	0.231	0.143	0.193
2021	0.236	0.184	0.142	0.157	0.159	0.185	0.113	0.163	0.203	0.163	0.128
2022	0.201	0.170	0.146	0.158	0.150	0.197	0.129	0.152	0.176	0.137	0.149
2023	0.153	0.152	0.125	0.153	0.142	0.216	0.132	0.143	0.184	0.104	0.135
2024	0.151	0.221	0.132	0.142	0.208	0.197	0.104	0.181	0.224	0.106	0.104
Mean	0.230	0.190	0.147	0.160	0.157	0.203	0.153	0.169	0.193	0.128	0.157

Table 8: Firm volatility by industry and year

	Country (%)	**Industry (%)**
Australia	57.9	42.1
Belgium	35.4	64.6
Canada	47.5	52.5
Switzerland	72.7	27.3
China	29.2	70.8
Germany	32.9	67.1
Denmark	38.5	61.5
France	44.6	55.4
United Kingdom	62.8	37.2
Hong Kong	56.2	43.8
Ireland	44.2	55.8
Italy	63.7	36.3
Japan	28.9	71.1
Netherlands	42.0	58.0
United States	52.5	47.5
Mean	47.2	52.8

Table 9: Firm volatility attributed to country and industry effect

same weight in explaining the stock volatility. This finding is in line with Timmermann and Catão (2003). They found that since the late 1990s, the industry factor accounts for about one-third of total systematic variance in stock returns. If earlier data is included, industry contribution reduces to only 10%. Another important finding from this table is that, the gap between country influence and industry influence is narrowing down. As of 2024, the two ratios are 49.8% and 50.2%. A possible reason is globalization. At the beginning of this century China joined WTO. The world is more and more connected. Firms from one country can have large business in other countries, and it is hard to distinguish which one is more important. If a firm have business in another country, all of its income will be recorded in its headquarter country. Therefore the country effect of the other countries where it have business is not taken into consideration in the analysis.

Year	**Country (%)**	**Industry (%)**
2005	52.1	47.9
2006	43.1	56.9
2007	43.6	56.4
2008	45.1	54.9
2009	44.9	55.1
2010	44.7	55.3
2011	44.0	56.0
2012	43.9	56.1
2013	44.2	55.8
2014	43.1	56.9
2015	44.7	55.3
2016	44.3	55.7
2017	43.2	56.8
2018	45.6	54.4
2019	44.8	55.2
2020	43.8	56.2
2021	44.5	55.5
2022	43.7	56.3
2023	43.5	56.5
2024	49.8	50.2
Mean	**44.4**	**55.6**

Table 10: Firm volatility attributed to country and industry effect across years

4.3 Robustness check

Since the country effect is very strong in the last century, and it is still prominent in the result paper, it is important to figure out if there is any other factors on this. Exchange rate was used to calculate the global company market capitalization at the beginning of this research. This could be a influence on the country effect, as the float of exchange rate can greatly affect import, export and investment from foreign countries. It may indirectly be taken as part of country effect in the previous regression and volatility decomposition. Hence the next step is going to follow Heston and Rouwenhorst (1994) model.

Table 11 shows the regression result of currency exchange rate percentage change on the country effects. Half of the coefficients are not significant, so we can not reject the hypothesis that exchange rate hardly have effect on the country effects. Moreover, for those coefficients which are significant, the effect is also nearly zero. This implies that change of exchange

Country	Coefficient	Std. Error	p-value
Belgium	0.000	0.000	0.238
Canada	0.001***	0.000	0.000
Switzerland	0.002***	0.000	0.000
China	0.000**	0.000	0.022
Germany	0.000**	0.000	0.030
Denmark	0.000	0.000	0.441
France	0.000	0.000	0.054
United Kingdom	-0.001***	0.000	0.000
Hong Kong	0.000	0.000	0.112
Ireland	0.000	0.000	0.514
Italy	0.000	0.000	0.242
Japan	-0.000***	0.000	0.000
Netherlands	0.000	0.000	0.032
Constant	-0.000***	0.000	0.006

$^{***}p < 0.01$, $^{**}p < 0.05$

Table 11: Effect of exchange rate on country

rate over time only account for a small portion of country effect, and does not influence the results above.

5. Conclusion

This paper implemented a weighted regression model to explore the effect of different countries and industries on stock returns. It decomposes stock return into three parts: global return, industry return and country return. Then the weight of every industry and country is constructed and applied in the restrictions. After that, stock return volatility is also decomposed into industry part and country part to investigate how much the two effects moves the volatility. Then a robustness check was applied to ensure the validity of the previous results.

There are two important findings in this paper. First, this paper finds that most included countries and industries have significant influences on the stock price. Among them, all industry factors are significant. The only exceptions are Denmark and Hong Kong. This result is robust, with proof from the exchange rate change regression. The average global factor, which is common for all firms, is 2.5% in the past twenty years.

Second, the volatility decomposition provides more information regarding the stronger factor and the development of them. From early research of other scholars, country dynamics played a main role in driving the movement of stock returns, compared with industry dynamics. Starting from this century, the importance of country dynamics has reduced and the gap between the two factors are narrower. In 2024, the two factors are equally important in the aspect of volatility.

With globalization, it is believed that it will be difficult to categorize a company as in which country or which industry. Modern companies may have business in many countries and industries. Therefore, the country effect and industry effect are very likely to be equally important in the future. It is also good for investors to put money in various countries and industries for global risk diversification.

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